

Apache Tomcat and Jenkins Pipeline for MavenAnsibleWebApp1 along with Ansible

Step 1: create a Web app by typing the command **mvn archetype:generate** –
DgroupId=com.example –DartifactId=MavenAnsibleWebApp1 –
DarchetypeArtifactId=maven-archetype-webapp –DinteractiveMode=false

```
samiksha@samiksha:~$ cd Devops6D
samiksha@samiksha:~/Devops6D$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenAnsibleWebApp01 -DarchetypeArtifactId
=maven-archetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/samiksha/Devops6D
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenAnsibleWebApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/samiksha/Devops6D/MyMavenAnsibleWebApp01
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.383 s
[INFO] Finished at: 2026-04-15T16:30:01+05:30
[INFO] -----
```

Step 2 : enter the MavenAnsibleWebApp1 , open the pom.xml file and paste the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.example</groupId>
```

```
<artifactId>MavenAnsibleWebApp1</artifactId>
```

```
<packaging>war</packaging>
```

```
<version>1.0-SNAPSHOT</version>
```

```
<name>MavenAnsibleWebApp1</name>
```

```
<url>http://maven.apache.org</url>
```

```
<dependencies>
```

```
<dependency>
```

```
<groupId>junit</groupId>
```

```
<artifactId>junit</artifactId>
```

```
<version>3.8.1</version>
```

```
<scope>test</scope>
```

```

</dependency>
<!-- Servlet API for Java EE -->
<dependency>
<groupId>javax.servlet</groupId>
<artifactId>javax.servlet-api</artifactId>
<version>4.0.1</version>
<scope>provided</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-war-plugin</artifactId>
<version>3.4.0</version>
<!-- Update to latest stable version -->
</plugin>
</plugins>
<finalName>MavenAnsibleWebApp1</finalName>
<!-- create a web page with the name MavenAnsibleWebApp, if we want some other name
we can chage it -->
</build>
</project>

```

Step 3: create Jenkinsfile by typing the command **gedit Jenkinsfile** and paste the code

```
pipeline {
```

```
    agent any // Use any available agent
```

```
    environment {
```

```
        LANG = 'en_US.UTF-8'
```

```
        LC_ALL = 'en_US.UTF-8'
```

```
    } // this has to be added only if you get an error saying UTF required is 8 but showing in
    ISO00009    tools {
```

```

    maven 'Maven' // Ensure this matches the name configured in Jenkins
  }
  stages {
    stage('Checkout') {
      steps {
        git branch: 'master', url:
'https://github.com/ShruthiBGowda/MavenAnsibleWebApp1-CICD.git'
      }
    }
    stage('Build') {
      steps {
        sh 'mvn clean package' // Run Maven build
      }
    }
    stage('Archive') {
      steps {
        archiveArtifacts artifacts: 'target/*.war', fingerprint:true
      }
    }
    stage('Deploy') {
      steps {
        sh 'mvn clean package'
        sh 'ansible-playbook ansible/playbook.yml -i ansible/hosts.ini'
      }
    }
  }
}

```

Step 4: create a new directory by typing the command **mkdir ansible** . Inside ansible create the hosts.ini file and playbook.yml file.

Paste the following code in **hosts.ini**

```
[web]
```

```
localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3
```

and paste this code in **playbook.yml**

```
- name: Deploy to Tomcat
```

```
  hosts: web
```

```
  become: yes
```

```
  tasks:
```

- name: Copy WAR file to Tomcat

copy:

src: "/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1-CICD/target/MavenAnsibleWebApp1.war"

dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"

remote_src: no

- name: Stop Tomcat

shell: /opt/tomcat/bin/shutdown.sh

ignore_errors: yes

- name: Start Tomcat

shell: /opt/tomcat/bin/startup.sh

```
samiksha@samiksha: ~/DevOps6D/MavenAnsibleWebApp1$ tree
```

```
.
├── Jenkinsfile
├── ansible
│   ├── hosts.ini
│   └── playbook.yml
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── WEB-INF
│           │   └── web.xml
│           └── index.jsp
```

7 directories, 6 files

Commit everything and push it to GitHub

Step 5: Go to Jenkins Dashboard click on new item and give the name MavenAnsibleWebApp1-CICD and select pipeline. In the configure select pipeline with scm paste the GitHub url and select git credentials apply and save .

Jenkins / MavenAnsibleWebApp1-C... / Configure

Configure

General Triggers Pipeline Advanced

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/Samiksha201104/MavenAnsibleWebApp1.git

Credentials ?

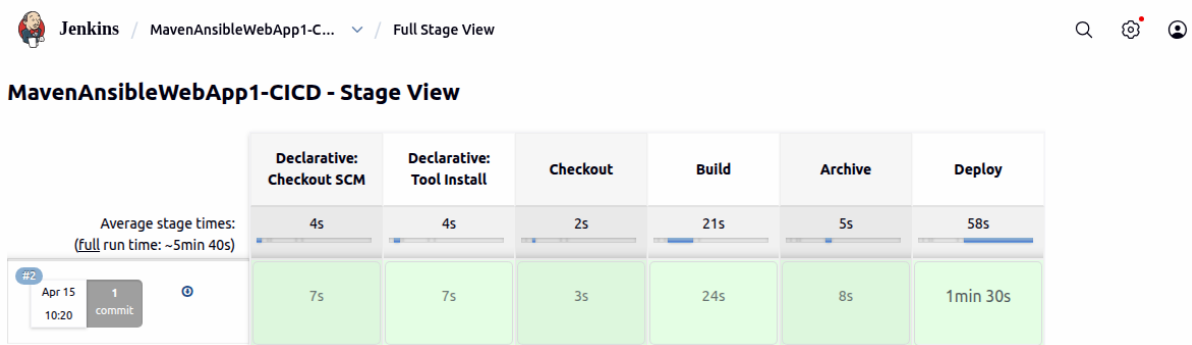
GitHub Credentials

+ Add

Advanced

Save Apply

Step 6: Go back to dashboard and build MavenAnsibleWebApp1-CICD



Step 7: Start Tomcat by typing the command **sh startup.sh**

```
samiksha@samiksha:~$ sudo -i
[sudo] password for samiksha:
root@samiksha:~# cd /
root@samiksha:~# cd opt
root@samiksha:/opt# cd tomcat/
root@samiksha:/opt/tomcat# cd bin
root@samiksha:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@samiksha:/opt/tomcat/bin#
```

Step 8: Login to Manager app using the credentials given and select the path /MavenAnsibleWebApp1



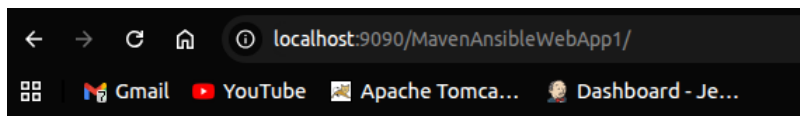
Tomcat Web Application Manager

Message: OK

Manager			
List Applications	HTML Manager Help	Manager Help	Server Status

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MavenAnsibleWebApp1	None specified	Archetype Created Web Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MymavenWebApp01	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

We will get hello world in the browser



Hello World!